

Gaurav Malik

Senior Backend Engineer | Node.js | Microservices | Distributed Systems

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PROFESSIONAL SUMMARY

Results-driven Backend Developer with **4.5+ years of experience** in designing **scalable, high-performance backend systems** using **Node.js, TypeScript, and Express.js**. Expertise in **System Design, Microservices Architecture, and Database Management (SQL & NoSQL)**. Proven ability to **optimize backend performance** and implement **secure, maintainable solutions**. Adept at building **robust APIs** and driving **fintech software initiatives**.

TECHNICAL SKILLS

- **Programming Languages:** JavaScript (ES6+), TypeScript, Node.js, Python
- **Frameworks:** Express.js, REST API Development, SOAP API Integration, GraphQL
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch
- **Cloud & Platforms:** AWS (S3, SQS, SNS), Docker, CI/CD Pipelines
- **Tools & Version Control:** Git, GitHub, Bitbucket, JFrog, Vault, Postman, VS Code, Jenkins
- **Methodologies & Concepts:** Microservices Architecture, System Design, Design Patterns, Data Structures & Algorithms, Computer Networks, Agile/Scrum
- **Domain Expertise:** Fintech Software Development, Secure Transactions, High-Performance APIs

PROFESSIONAL EXPERIENCE (4.5+ Years)

Senior Software Development Engineer (SSDE) | *Mar 2023 - Present*

Freecharge | *Remote*

- **Designed and developed large-scale fintech systems** including a real-time Digital Gold transaction platform (Buy, Sell, Delivery, SIPs) and National Pension System (NPS) services from scratch, serving **1.5M + customers** in collaboration with vendors like **Kfintech & SafeGold**.
- **Contributed to high-volume payment services** (debit, credit, and exception handling), ensuring **secure, reliable, and fault-tolerant transactions**.
- **Currently contributing to the Mutual Fund platform** from the ground up, developing **Buy, Sell, SIP, SWP, and KYC journey** workflows, integrating high-volume payment services, **implementing real-time NAV/fund detail updates**, and building automated reconciliation and operational reporting through cron-based processes.
- **Utilized AI-Driven Development Lifecycle (AIDLC) tools** for **architecture brainstorming, code optimization, test case generation, and technical documentation**, reducing development effort and accelerating feature delivery.

System Engineer | *Apr 2021 - Sep 2022*

Tata Consultancy Services | *Remote*

- **Built RESTful microservices** using **Node.js and Express.js**, ensuring **scalability, reliability, and seamless system integration**.
- **Designed and deployed high-performance server-side components**, collaborating with front-end teams for **end-to-end delivery**.
- **Developed API performance optimizations** using **asynchronous processing, caching (Redis), and connection pooling**, reducing latency and improving throughput.

PROJECTS

Gold Loan Renewal Journey | **Axis Bank**

- **Designed and developed a fully automated Gold Loan Renewal platform** for Axis Bank, reducing **loan disbursement time to under 3 minutes** through **process optimization and automation**.
- **Eliminated manual intervention** by creating a **digital, hassle-free customer journey** with end-to-end automation.
- **Managed payment exceptions** and implemented **Cron job-based automation** for **loan disbursement and document generation**.
- **Integrated Finacle and ESB services** to ensure **customer record validation and seamless data synchronization** via automated jobs.
- **Tech Stack & Skills:** Node.js, Express.js, REST APIs, MongoDB, Redis, Cron Jobs, Finacle Integration, ESB, Microservices Architecture, Payment Exception Handling, AWS SQS/SNS, S3

Digital Gold (Lump Sum Buy/Sell + SIP) | Axis Bank

- Led **migration from PHP/.NET SOAP monolith to RESTful Microservices**, improving **UI, scalability, and business agility**.
- Processed **high-volume daily buy/sell transactions**, reducing **failure rates by 30%** and improving **system accuracy**.
- **Optimized Digital Gold SIP cron jobs**, reducing execution time from **10 hours to 15 minutes (90% improvement)** through **workflow optimization and distributed processing**.
- Built **automated reconciliation and refund systems**, enabling **data-driven decisions**, eliminating **stakeholder dependency**, and ensuring **efficient processing**.
- Resolved **SIP duplication issues** and managed **complex credit/debit scenarios**, ensuring **reliable and seamless transactions**.
- **Tech Stack & Skills:** Node.js, Express.js, REST APIs, SOAP APIs, MongoDB, Redis, Cron Jobs, Finacle Integration, ESB, Microservices Architecture, Payment Exception Handling, AWS SQS/SNS, S3

National Pension Scheme (NPS) | Axis Bank

- **Built the NPS platform from scratch**, integrating with **KFintech** as the backend service provider.
- Developed **Lump Sum Investment** functionality for **Tier 1 and Tier 2 funds**, enabling **seamless user investments**.
- Implemented **Monthly Auto Investment (SIP)** with **automated contributions** based on user preferences.
- Designed **secure, compliant APIs** and **optimized transaction workflows**, improving **performance** and reducing **processing time**.
- **Tech Stack & Skills:** Node.js, Express.js, REST APIs, MongoDB, Redis, Cron Jobs, Finacle Integration, ESB, Microservices Architecture, Payment Exception Handling, AWS SQS/SNS

Mutual Fund - Notification System | Axis Bank

- Designed a **distributed notification service** for Mutual Fund operations, **processing Buy/Sell/SIP events** through **Kafka-based asynchronous pipelines**.
- Designed and **developed a centralized notification service** for the Mutual Fund ecosystem, enabling real-time customer communications across transactions, SIPs, KYC updates, order status changes, and operational events through **scalable event-driven architecture**.
- Implemented PostgreSQL-backed notification storage and Redis-based unread counters to **optimize user notification feeds and reduce database load**.
- Enhanced system **scalability** and **reliability** through **event-driven architecture, background workers, retry mechanisms, and channel-specific notification services**.

EDUCATION

Bachelor of Technology in Computer Science & Engineering | *Sep 2020*

Future Institute of Engineering & Technology | *Bareilly, UP*

ACHIEVEMENTS

- Recognized for independently driving project success and consistently delivering on time.
- Awarded the Rockstar Performance Award for developing a highly scalable payment processing system.